First Steps with .NET

A quick beginner-friendly guide to getting started with .NET, writing your first app, and learning the core concepts.

1. What is .NET?

- .NET is a development platform for building web apps, desktop apps, mobile apps, APIs, cloud services, and more.
- You can write apps in C#, F#, or Visual Basic, but C# is the most common choice for beginners.

2. Install the SDK

- Download and install the latest .NET SDK from the official .NET website.
- After installation, open a terminal and run: dotnet --version

3. Create your first app

- Open a terminal.
- Run: dotnet new console -n HelloDotNet
- Go into the folder: cd HelloDotNet
- Run the app: dotnet run

4. Understand the project files

- Program.cs: the entry point of the app.
- HelloDotNet.csproj: project configuration and package references.
- bin/ and obj/: build output and temporary files.

5. Learn the basics of C#

- Variables and types: string, int, bool, double.
- Conditions: if / else.
- Loops: for, while, foreach.
- Methods: reusable blocks of code.
- Classes and objects: how you structure real applications.

6. Use packages

- You can add libraries from NuGet, the .NET package manager.
- Example: dotnet add package PackageName

7. Build something small

- Try a calculator, to-do list, JSON reader, or a simple API.
- Small projects help you learn faster than only reading tutorials.

8. Next steps

- Learn async/await.
- Practice reading and writing JSON.
- Explore ASP.NET Core for web apps and APIs.
- Try Entity Framework Core if you need a database.

Tip: If you want to learn faster, build one small project at a time and keep using dotnet run, dotnet build, and dotnet test.